

The Sharp Slope-One Decay Constant

1 Definitions

Let

$$\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z}).$$

Define

$$\mathcal{A} = \left\{ f \in W^{1,\infty}(\mathbb{T}) : \int_{\mathbb{T}} f \, dx = 0, \|f_x\|_{L^\infty} \leq 1 \right\}. \quad (1)$$

Then

$$\lambda_* = \inf_{\substack{f \in \mathcal{A} \\ f \neq 0}} \frac{\int_{\mathbb{T}} f_x \arctan(f_x) \, dx}{\int_{\mathbb{T}} f^2 \, dx}. \quad (2)$$

2 Best Known Result

For $|p| \leq 1$,

$$p \arctan p \geq \frac{\pi}{4} p^2. \quad (3)$$

Together with the sharp mean-zero Poincaré inequality on $\mathbb{T}$, this gives

$$\lambda_* \geq \frac{\pi}{4}. \quad (4)$$

Testing with $f(x) = \sin x$ gives

$$\lambda_* \leq \frac{\int_{\mathbb{T}} \cos x \arctan(\cos x) \, dx}{\int_{\mathbb{T}} \sin^2 x \, dx} = 2(\sqrt{2} - 1). \quad (5)$$

Hence the elementary certified bracket is

$$\frac{\pi}{4} \leq \lambda_* \leq 2(\sqrt{2} - 1). \quad (6)$$

The exact value, or a certified enclosure meeting the precision requested below, is not currently supplied by these estimates.

3 Problem Formalization

Determine λ_* as accurately as possible.